

OLIVIA A. M. MARTIN

Curriculum Vitae

Email — Website — LinkedIn — Google Scholar

RESEARCH INTERESTS & EXPERTISE

- Water-mineral interfaces, H-bonding structure and dynamics, computational spectroscopy
- Membrane-mineral & membrane-water interfaces, prebiotic chemistry, origins of life, thermodynamics of membrane formation
- Machine learning, statistical mechanics
- Computational & Theoretical Methods: Molecular dynamics (MD) simulations, path integral MD, machine learning, umbrella sampling, indirect umbrella sampling (INDUS), statistical mechanics, nuclear quantum effects
- Software & Hardware Skills: GROMACS, LAMMPS, i-PI, PLUMED, high performance computing (HPC), Python (NumPy, SciPy, PyTorch)

EDUCATION

Ph.D. in Chemistry, Physical Chemistry
Advisor: Dr. Richard C. Remsing

Expected May 2027
Rutgers University, Piscataway, NJ

B.S. in Chemistry, *cum laude*
Advisor: Dr. Nathan C. Eddingsaas
Minor: Environmental Sciences

May 2022
Rochester Institute of Technology, Henrietta, NY

RESEARCH EXPERIENCE

Graduate Student Researcher

June 2022 - Present

Advisor: *Dr. Richard C. Remsing*

Department of Chemistry and Chemical Biology, Rutgers University, Piscataway, NJ

Emerson Summer Undergraduate Research Fellowship

May - August 2021

Advisor: *Dr. Nathan C. Eddingsaas*

College of Science, Rochester Institute of Technology, Henrietta, NY

Undergraduate Student Researcher

August 2018 - May 2022

Advisor: *Dr. Nathan C. Eddingsaas*

Department of Chemistry and Materials Science, Rochester Institute of Technology, Henrietta, NY

PUBLICATIONS

Peer-Reviewed Publications

1. "A bellwether for microplastic in wetland catchments in the Great Lakes region" Dominique S. Gilbert, Brett A. Hayhurst, Sarah Grubisich, Nick Schneider, **Olivia Martin**, Christopher DeNyse, Kristina M. Chomiak, Anna Christina Tyler, Nathan C. Eddingsaas. *Journal of Great Lakes Research*, **2024**, 50, 102411
DOI: 10.1016/j.jglr.2024.102411

CURRENT AND PAST FUNDING

External Grants

Funded

Proposal Title: Astrobiology Graduate Conference (ABGradCon) 2026

Funding Agency: National Aeronautics and Space Administration

Role Status: Collaborator

Percentage of Attributed Effort: 20

Total direct costs and annual budget: \$119,640

Coverage Period: June - December 2026

HONORS AND AWARDS

2022 Undergraduate Senior Chemistry Achievement Award, *American Institute of Chemists*
2021 Nathaniel Rochester Society Scholarship
2021 Good Citizenship Award, *Rochester Institute of Technology*
2020 Nathaniel Rochester Society Scholarship
2020 Good Citizenship Award, *Rochester Institute of Technology*
2020 Alumni Endowed Academic Scholarship, *College of Science, Rochester Institute of Technology*
2020 Outstanding Second Year Student Award, *Rochester Institute of Technology*

SELECTED TALKS AND PRESENTATIONS

Oral Presentations

7. "Molecular and Thermodynamic Insights into Water and Amphiphiles at the α -Alumina Interface", Water and Aqueous Solutions Gordon Research Seminar 2026, Holderness, NH
6. "Molecular and Thermodynamic Insights into Prebiotic Lipid Assembly", Astrobiology Science Conference 2026, Madison, WI
5. "Lipid Assembly Formation on Minerals: A Path to Protocells", Astrobiology Graduate Conference 2025, Boulder, CO
4. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Astrobiology Graduate Conference 2024, Ithaca, NY
3. "Interfacial Structure and Vibrational Relaxation of Aqueous Solutions near Alumina", American Chemical Society Mid-Atlantic Regional Meeting (MARM) 2024, State College, PA
2. "Analytical Methods and Processes of Environmental Microplastic", RIT Undergraduate Research Symposium 2021, Rochester, NY
1. "The chemical and physical changes to plastics aged in surface and benthic freshwater aquatic systems", State of Lake Ontario Conference 2021, Rochester, NY

Poster Presentations

9. "Molecular and Thermodynamic Insights into Water and Amphiphiles at the α -Alumina Interface", Water and Aqueous Solutions Gordon Research Conference 2026, Holderness, NH
8. "Thermodynamic Characteristics of Water and Amphiphiles at the Alumina Interface", Rutgers Chemical Physics Symposium 2026, Piscataway, NJ
7. "Structure and Dynamics of Water and Amphiphiles at the α -Alumina Surface", Chemistry and Physics of Liquids Gordon Research Conference 2025, Holderness, NH
6. "Structure and Dynamics of Water and Amphiphiles at the α -Alumina Surface", Chemistry and Physics of Liquids Gordon Research Seminar 2025, Holderness, NH
5. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Chemistry in Solution and at Interfaces Workshop: Deep Modeling for Molecular Simulations Workshop 2024, Princeton, NJ
4. "Fatty Acid Assembly at Water-Mineral Interfaces, Water and Aqueous Solutions", Water and Aqueous Solutions Gordon Research Conference 2024, Holderness, NH
3. "Fatty Acid Assembly at Water-Mineral Interfaces, Water and Aqueous Solutions", Water and Aqueous Solutions Gordon Research Seminar 2024, Holderness, NH
2. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Astrobiology Science Conference 2024, Providence, RI
1. "Microplastic Pollution in the Benthos of Rural and Urban Tributaries of Southern Lake Ontario", Joint Aquatic Sciences Meeting 2022, Grand Rapids, MI

TEACHING AND ADVISING

Graduate Teaching Assistant Organic Chemistry I Lab	2024 - Present Rutgers University, Piscataway, NJ
Graduate Mentor Remsing Lab	2024 - 2025 Rutgers University, Piscataway, NJ
Tutor Organic Chemistry I and II	2022 - 2023 Rutgers University, Piscataway, NJ
Undergraduate Mentor College of Science	2021 - 2022 Rochester Institute of Technology, Henrietta, NY
Undergraduate Teaching Assistant General and Analytical Chemistry Lab	2020 - 2022 Rochester Institute of Technology, Henrietta, NY
Learning Assistant General Chemistry I and II	2020 - 2021 Rochester Institute of Technology, Henrietta, NY

PROFESSIONAL SERVICE AND LEADERSHIP

Representative Rutgers Chemical Physics Early-Career Campus Committee	20256 - Present Rutgers University, Piscataway, NJ
Co-chair Chemistry and Physics of Liquids Gordon Research Seminar 2027	2025 - Present Holderness, NH
Directory of Fundraising and Communications Astrobiology Graduate Conference 2026	2025 - Present University of Arizona, Tucson, AZ
Discussion Leader Chemistry and Physics of Liquids Gordon Research Seminar 2025	2025 Holderness, NH
Lab Manager Remsing Lab	2024 - Present Rutgers University, Piscataway, NJ
Director of Outreach Chemistry and Chemical Biology Graduate Student Organization	2023 - 2025 Rutgers University, Piscataway, NJ
Executive Board Treasurer, Supporting Parents and Caregivers at Rutgers Organization	2023 - 2025 Rutgers University, Piscataway, NJ
Volunteer Women in Science	2019 - 2022 Rochester Institute of Technology, Henrietta, NY
Representative Young Chemists Committee, American Chemical Society	2020 - 2021 Rochester Institute of Technology, Henrietta, NY
President American Chemical Society	2019 - 2021 Rochester Institute of Technology, Henrietta, NY
Student Ambassador College of Science Student Advisory Board	2018 - 2021 Rochester Institute of Technology, Henrietta, NY
Volunteer Rochester Museum and Science Center	2020 Rochester, NY
Event Coordinator American Chemical Society	2019 Rochester Institute of Technology, Henrietta, NY
Chair	2019

REFERENCES

Richard C. Remsing, PhD

Associate Professor

Department of Chemistry and Chemical Biology, Rutgers University

123 Bevier Rd, Piscataway, NJ 08854

Phone: (848) 445-3245 | Email: rick.remsing@rutgers.edu

Relationship: Primary PhD Advisor

Lu Wang, PhD

Associate Professor

Department of Chemistry and Chemical Biology, Rutgers University

123 Bevier Rd, Piscataway, NJ 08854

Phone: (848) 445-4555 | Email: lwang@chem.rutgers.edu

Relationship: Thesis Committee Member

Paul G. Falkowski, PhD

Board of Governors Professor; Director, Rutgers Energy Institute

Department of Chemistry and Chemical Biology, Rutgers University

123 Bevier Rd, Piscataway, NJ 08854

Phone: (732) 932-6555 ext 370 | Email: falko@imcs.rutgers.edu

Relationship: Thesis Committee Member

Harender Singh Dhattarwal, PhD

Postdoctoral Research Associate

Department of Chemical and Biomolecular Engineering, Cornell University

113 Ho Plaza, Ithaca, NY 14853

Phone: (908) 210-4331 | Email: harender.dhattarwal@rutgers.edu

Relationship: Mentor; Postdoctoral Research Associate, Remsing Lab